RestoERP – Software Requirements Specification (SRS)

Version 1.1 · Scalable platform for small, medium and large restaurant operations · Multi-site, workspace-based · Includes Waiter Commission, Customer Portal and Order Routing

Revision note (v1.1): v1.0 content is retained, except that the Accounting/Finance and HR (rostering, attendance, payroll) modules are removed from this version and deferred to a later release. New in v1.1: Section 5 (Waiter Management, Commission, Customer Portal and Order Routing), extended data model, business rules, acceptance criteria, permission matrix and API appendix.

1. Introduction

1.1 Purpose

This document specifies the requirements for RestoERP, a cloud-first, modular Enterprise Resource Planning platform for restaurants. It must serve a single-outlet café, a growing multi-branch chain, and a large franchise or central-kitchen group using the same core product, activated by tier and modules.

1.2 Scope

RestoERP covers the full restaurant value chain: front-of-house sales, kitchen operations, inventory and recipe costing, procurement, central production, customer engagement, analytics and head-office control. It also covers **waiter commission management, per-waiter customer ordering links, a restaurant-scoped customer ordering portal, order routing and manager reconciliation**. Each physical site receives its own Site Workspace, while owners and head office see consolidated HQ Workspaces.

1.3 Definitions

Term	Meaning
Tenant	One customer company (owner/group)
Brand	A restaurant concept under a tenant
Site	A physical outlet, kitchen, warehouse or kiosk

Term	Meaning
Workspace	Role-based, site-scoped working environment with its own dashboards, tasks and permissions
POS	Point of Sale
KDS	Kitchen Display System
BOM / Recipe	Ingredient list with quantities and yield
CK	Central Kitchen
PAR level	Target stock level per item per site
COGS	Cost of goods sold
Customer Portal	Public, restaurant-scoped web ordering interface (installable PWA where supported)
Waiter link	Unique, revocable public URL that attributes a customer's order to a specific waiter; never an authentication credential
Commission ledger	Append-only record of commission earnings, approvals, payouts and reversals
Policy snapshot	Copy of the commission rate and basis stored on each ledger entry at posting time

1.4 Intended audience

Product owners, business analysts, architects, developers, QA, implementation consultants and restaurant management.

1.5 Product vision

One system that is simple enough for a 10-seat shop to start in a day, and deep enough for a 500-site chain to control cost, quality and growth.

2. Overall Description

2.1 Product tiers

Capability	Starter (1–2 sites)	Growth (3–25 sites)	Enterprise (25+ sites, franchise, CK)
POS, tables, KDS, basic reports	✔	✔	✔
Inventory and recipes	Basic	Full	Full + forecasting
Procurement	Simple PO	Approval flows	Contracts, tenders, 3-way match
Multi-site HQ	–	✔	✔ + franchise royalty engine
Central kitchen	–	Optional	✔
BI and AI forecasting	Standard reports	Dashboards	Custom BI, ML forecasting
API / integrations	Limited	Standard	Full, webhooks, SSO
Customer portal and waiter links	Basic portal	Waiter links + commission	Full routing rules, branch/category commission rules

Modules are feature-flagged per tenant so customers pay for and see only what they need. Upgrading never requires data migration.

2.2 User classes

- **Owner / Executive** – cross-site performance, profit, cash, waiter comparison.
- **Regional / Area Manager** – compares and controls a cluster of sites.
- **Site (General) Manager** – runs one outlet daily; configures commission, reconciles waiters.
- **Cashier / Waiter** – orders, payments, tables; waiters also share their ordering link and view their own earnings.
- **Kitchen Chef / Line Cook** – tickets, prep, waste.
- **Storekeeper** – receiving, issuing, stock counts.

- **Purchasing Officer** – suppliers, orders, price control.
- **Marketing / CRM Officer** – loyalty, promotions.
- **Franchisee** – own-site view only.
- **Auditor (read-only)** and **System Administrator**.
- **Customer** – ordering (including via a waiter link), reservations, loyalty (web/app).
- **Supplier** – portal for orders and invoices.

2.3 Operating environment

- Cloud SaaS (multi-tenant) with optional private/on-premise deployment for enterprise.
- Clients: Web (responsive PWA), Android/iOS tablet POS, handheld waiter app, KDS screens, customer web/app, supplier portal.
- Hardware: receipt/kitchen printers (ESC/POS, network/USB/Bluetooth), cash drawers, barcode scanners, scales, card/mobile-money terminals, fiscal devices, label printers.

2.4 Design constraints

- Must keep trading when the internet fails (offline-first POS).
- Must support multi-currency, multi-language (incl. RTL), multi-tax, local fiscal regulations.
- Must run acceptably on low-cost tablets and low bandwidth.

2.5 Assumptions

Sites have at least intermittent connectivity; each site has a designated manager; master data (menu, items, suppliers) can be shared across sites but overridden locally.

3. Organization and Workspace Model (core of the design)

3.1 Hierarchy

```
Tenant (Company Group)
└─ Legal Entity (tax registration, currency)
   └─ Brand
      └─ Region / Area
         └─ Site (Restaurant, Kiosk, Cloud Kitchen, Central Kitchen,
Warehouse)
            └─ Station / Terminal (POS 1, Bar, Grill, Pass, Store)
```

Every transaction carries tenant, entity, brand, region, site, station, user and timestamp.

3.2 Workspace types

1. **Group HQ Workspace** – owners/executives: consolidated KPIs, P&L by site, cash position, alerts, approvals.
2. **Regional Workspace** – benchmarking, audit checklists, stock transfers, regional promotions.
3. **Site Workspace** – one per outlet; the daily operating console described in Section 6.
4. **Central Kitchen / Warehouse Workspace** – production plans, internal orders, dispatch.
5. **Franchisee Workspace** – isolated view of own sites plus royalty statements.
6. **Procurement Workspace** – cross-site functional views.

3.3 Workspace rules

- **WS-01** Creating a site automatically provisions a Site Workspace from a template (roles, dashboards, printers, tax, menu, floor plan).
- **WS-02** Each workspace has its own dashboards, task inbox, notifications, calendar, and daily checklists.
- **WS-03** Data isolation: a user sees only workspaces and data assigned to them; cross-site visibility needs explicit grants.
- **WS-04** Inheritance: menu, prices, recipes, taxes, policies are defined at brand/group level and may be overridden per site with audit trail.
- **WS-05** Site templates and cloning allow opening a new outlet in under one day.
- **WS-06** Site lifecycle states: Draft → Pilot → Active → Temporarily Closed → Archived, with data retention.
- **WS-07** Users can belong to multiple workspaces with different roles (e.g., manager of Site A, viewer at Site B) and switch via a workspace switcher.
- **WS-08** Day-end closing per site locks the business day, reconciles cash and stock, and pushes results to HQ.

4. Functional Requirements (by module)

Priority: M = Must, S = Should, C = Could.

4.1 Point of Sale (POS)

ID	Requirement	Pri
POS-01	Dine-in, takeaway, delivery, drive-through, bar tab, kiosk and self-order modes	M
POS-02	Table map with drag-and-drop floor plans, merge/split/transfer tables and checks	M
POS-03	Modifiers, combos, courses, allergen flags, special notes, seat numbers	M
POS-04	Split bills (by item, seat, amount, percentage); partial and mixed payments	M
POS-05	Payments: cash, card, mobile money, QR, vouchers, gift cards, loyalty points, house accounts, tips	M
POS-06	Discounts, promotions, happy hour, coupons with role-based limits and approval	M
POS-07	Void/refund/comp with reason codes, manager PIN, full audit	M
POS-08	Offline mode: continue selling, printing and payments (cash) for at least 72 hours; automatic conflict-free sync	M
POS-09	Fiscal receipt integration (e.g., EFD/VFD/e-invoicing per country), tax-inclusive and exclusive pricing	M
POS-10	Shift open/close, cash drops, blind cash count, over/short report	M
POS-11	Price scheduling by time, channel, site, or customer group	S
POS-12	Waiter handheld ordering with fire-to-kitchen	M
POS-13	Customer-facing display and QR self-ordering at table	S
POS-14	Voice/AI upsell suggestions at order entry	C

ID	Requirement	Pri
POS-15	Every order records its source, creator and assigned waiter separately (see Section 5)	M

4.2 Kitchen Operations and KDS

ID	Requirement	Pri
KDS-01	Route items to stations (grill, fry, cold, bar, pastry) by rules	M
KDS-02	Ticket timers, color-coded urgency, bump, recall, expo/pass screen	M
KDS-03	Course firing and hold	M
KDS-04	Printer fallback if a KDS screen fails	M
KDS-05	Prep lists computed from forecast and open orders	S
KDS-06	86-ing (out-of-stock) propagated live to POS, online and kiosks	M
KDS-07	Kitchen performance analytics (ticket time per station and item)	S
KDS-08	Customer-portal and waiter-link orders reach the kitchen queue automatically with no waiter re-entry	M

4.3 Reservations, Waitlist and Events

- **RES-01 (M)** Online and phone reservations with table assignment, capacity rules, turn-time estimates.
- **RES-02 (M)** Waitlist with SMS/WhatsApp notification.
- **RES-03 (S)** Deposits and no-show fees.
- **RES-04 (S)** Banquet/event management: quotes, packages, deposits, BEO sheets, headcount-based production.

4.4 Online Ordering, Delivery and Aggregators

- **ONL-01 (M)** Branded website/app ordering with live menu, availability and site selection (detailed in Section 5.3).
- **ONL-02 (M)** Aggregator integration (marketplaces, local delivery apps): orders injected to POS/KDS, menu and availability sync, commission tracking.
- **ONL-03 (M)** Delivery management: zones, fees, driver assignment, live tracking, proof of delivery.
- **ONL-04 (S)** Cloud/ghost kitchen support: virtual brands sharing one kitchen.
- **ONL-05 (M)** Per-waiter ordering links and order routing rules (Section 5.3–5.4).

4.5 Menu, Pricing and Recipe Management

- **MNU-01 (M)** Central menu with categories, items, variants, modifiers, combos, allergens, nutrition, images, multi-language names.
- **MNU-02 (M)** Recipe/BOM with sub-recipes, yields, wastage %, unit conversions.
- **MNU-03 (M)** Real-time recipe costing from latest purchase cost; margin and food-cost % per item.
- **MNU-04 (S)** Menu engineering matrix (stars, plowhorses, puzzles, dogs).
- **MNU-05 (S)** Menu versioning, scheduled publishing, approval workflow.
- **MNU-06 (M)** Price lists per site, channel, and customer group.

4.6 Inventory and Warehouse

ID	Requirement	Pri
INV-01	Item master: stock, non-stock, prep, packaging, assets; multiple UoM; barcodes/SKUs	M
INV-02	Multi-location stock (store, kitchen, bar, freezer, CK, warehouse)	M
INV-03	Receiving against PO with quality check, temperature log, partial delivery	M
INV-04	Automatic depletion by recipe on sale (theoretical usage)	M
INV-05	Stock count: full, cycle, blind; variance (actual vs theoretical) with approval	M

ID	Requirement	Pri
INV-06	Inter-site and inter-location transfers with in-transit tracking	M
INV-07	Batch/lot and expiry tracking, FEFO issuing, expiry alerts	M
INV-08	Waste and spoilage logging with reason, photo, cost	M
INV-09	PAR levels, reorder points, auto-suggested requisitions	M
INV-10	Costing methods: weighted average, FIFO, last cost, standard cost	M
INV-11	Mobile stock counting with barcode scanning and scale integration	S
INV-12	Demand forecasting using sales history, weather, events, seasonality	S

4.7 Procurement and Supplier Management

- **PRC-01 (M)** Supplier master with catalogs, contracts, lead times, price history, ratings.
- **PRC-02 (M)** Requisition → RFQ → PO → Goods Receipt → Invoice (3-way match).
- **PRC-03 (M)** Configurable approval chains by amount, category and site.
- **PRC-04 (S)** Supplier portal: view POs, confirm, upload invoices, track payments.
- **PRC-05 (S)** Price variance alerts and best-price recommendation.
- **PRC-06 (C)** EDI and e-ordering to major suppliers.

4.8 Central Kitchen and Production

- **CK-01 (M)** Internal orders from sites to CK; consolidated production plan.
- **CK-02 (M)** Production orders with BOM, yield, batch numbers, labor and overhead costing.
- **CK-03 (M)** Dispatch, delivery notes, route planning, receiving confirmation at sites.
- **CK-04 (S)** HACCP records, label printing with date, batch and allergens.

4.9 Customer Relationship and Marketing

- **CRM-01 (M)** Customer profiles, visit history, preferences, allergies.
- **CRM-02 (S)** Loyalty: points, tiers, stamps, wallet, referral.
- **CRM-03 (S)** Campaigns by SMS, email, WhatsApp, push; segmentation; promo code tracking and ROI.
- **CRM-04 (S)** Feedback and reviews capture (QR, post-visit survey), alerts on low scores.
- **CRM-05 (C)** Gift cards and e-vouchers.

4.10 Reporting, Analytics and BI

- **RPT-01 (M)** Standard reports: sales by item/category/hour/channel/server, voids, discounts, payments, tax, shift, COGS, waste, variance, labor, supplier spend.
- **RPT-02 (M)** Role dashboards in every workspace (see Section 6) with drill-down from group to site to order.
- **RPT-03 (M)** Scheduled reports by email/WhatsApp; export to CSV/Excel/PDF.
- **RPT-04 (S)** Site benchmarking and ranking by KPI.
- **RPT-05 (S)** Self-service report builder and data warehouse export.
- **RPT-06 (C)** AI insights: anomaly/fraud detection, forecast, "why did sales drop" narrative.
- **RPT-07 (M)** Waiter reconciliation and commission reports (Section 5.7).

4.11 Quality, Safety and Compliance

- **QSC-01 (M)** Digital checklists (opening, closing, cleaning, safety) with photos and manager sign-off.
- **QSC-02 (M)** Temperature and HACCP logs (manual or IoT sensor).
- **QSC-03 (S)** Audit scheduler with scoring and corrective-action tracking.
- **QSC-04 (S)** Allergen and traceability reports (forward/backward by batch).

4.12 Asset and Maintenance

- **AST-01 (S)** Equipment register, maintenance schedules, breakdown tickets, vendor contacts, warranty.
- **AST-02 (C)** IoT monitoring for fridges and ovens.

4.13 Administration and Configuration

- **ADM-01 (M)** Tenant, entity, brand, region, site, station setup wizards.
- **ADM-02 (M)** Role-based access control with granular permissions, custom roles, field-level restrictions.
- **ADM-03 (M)** Approval workflow designer (discounts, POs, refunds, stock adjustments, expenses, commission payouts).
- **ADM-04 (M)** Audit log of every create/update/delete/approve action, tamper-evident.
- **ADM-05 (M)** Data import tools (menu, items, suppliers, opening stock, staff) with validation.
- **ADM-06 (S)** Notification center with rules (email, SMS, push, WhatsApp).
- **ADM-07 (S)** Localization packs: language, currency, tax, fiscal, payroll rules.

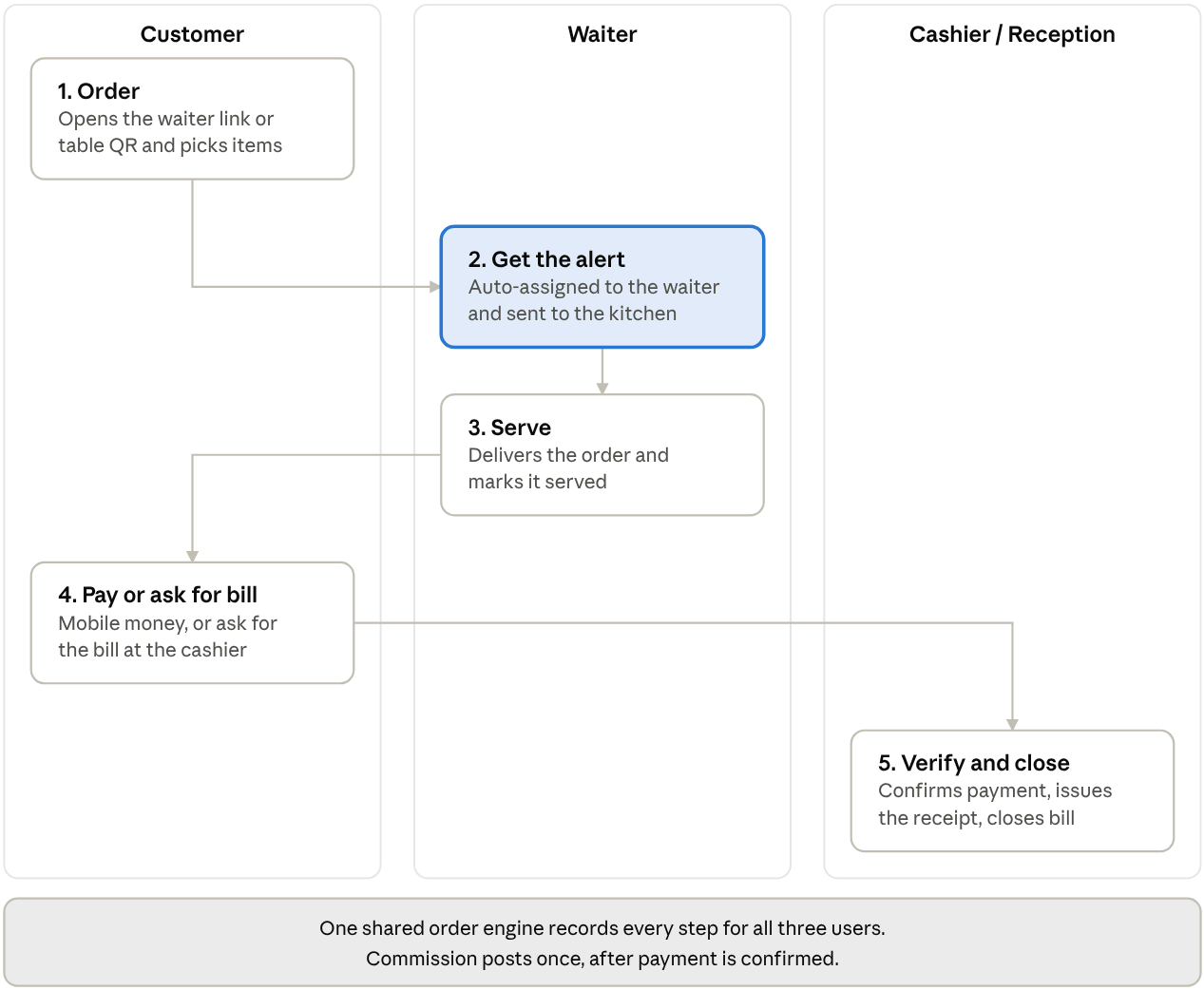
5. Waiter Management, Commission, Customer Portal and Order Routing

This section specifies how the owner or manager configures waiter commissions, how each waiter gets a personal customer-ordering link, how online orders are assigned to the correct waiter, and how every order, payment and commission is reconciled. It **extends the shared order engine**; it must not create a separate ordering system for each interface.

5.0 Core operating flow – Customer, Waiter and Cashier

Three people use the system every day, and they all work on one shared order record, so nobody types an order twice. Each step is one tap or one screen for the person doing it, and the next person is notified automatically.

One order moves from customer to waiter to cashier without re-entry



three-lane flow · 5 steps

The lifecycle of every order is the same whichever way it started: **Placed → Accepted by kitchen → Preparing → Ready → Served → Bill requested → Paid → Closed**. Cancellation is allowed before Served, with a reason and manager approval.

User	Their one screen	Main actions	Never has to
Customer (no login)	Restaurant portal, opened from a waiter link, table QR or general link	Browse menu, order, track status, pay or ask for the bill	Create an account, install an app, or know the waiter's name

User	Their one screen	Main actions	Never has to
Waiter (phone)	Waiter home: My Orders (live), New Order, Share My Ordering Link, My Earnings	Accept alerts, take walk-in orders, mark served, request bill, take payment where allowed	Re-enter orders for the kitchen, or calculate commission
Cashier / Reception (tablet or desktop)	Cashier desk: Bill Queue, New Order, Cash Summary	Verify payment, issue receipt, close bill, create orders for customers at the desk	Chase waiters for bills, or work out which waiter owns an order

ID	Requirement	Pri
FLW-01	Customer, waiter and cashier all act on one shared order record; every action is a status change on that record, never a second copy	M
FLW-02	Orders follow the single lifecycle above for every source (waiter link, table QR, general portal, walk-in, cashier-created)	M
FLW-03	Each status change notifies the next person automatically: waiter alert on new order, kitchen ticket, customer status page, cashier bill queue	M
FLW-04	The customer needs no account or app; one link opens one restaurant's menu and tracks that order	M
FLW-05	Each role sees only its own screen and the actions it needs (waiter home has 4 areas, cashier desk has 3); a user with several roles switches via the workspace switcher (WS-07)	M
FLW-06	A bill request from the customer or waiter appears instantly in the cashier's Bill Queue with table, waiter, items and amount due	M

ID	Requirement	Pri
FLW-07	A verified mobile-money payment closes the bill automatically; cash and other payments are recorded by the cashier or an authorized waiter, and each path posts the payment and commission exactly once	M
FLW-08	The cashier can create an order for a customer at the desk and assign or change the waiter afterwards, with the change logged	S
FLW-09	Waiter and customer screens are built mobile-first for low-bandwidth phones; the cashier desk is built for tablet and desktop	M

5.1 Tenant isolation principle

The customer portal is generated from each restaurant's own account. Every restaurant has its own menu, tables, staff, ordering link and settings. One restaurant must never be able to access another restaurant's data (see NFR-11, WS-03).

5.2 Commission configuration and ledger

The manager has a dedicated **Waiter Commission Settings** page.

ID	Requirement	Pri
WCM-01	Calculation methods: percentage per eligible order; fixed amount per eligible order; tiered commission based on sales targets (manager-configured tier table)	M
WCM-02	Commission basis: gross order total, or net eligible sales after discounts excluding tax where configured	M
WCM-03	Qualification rule per policy: only after payment is confirmed, or after the order is served and manager-approved	M

ID	Requirement	Pri
WCM-04	Default commission policy per restaurant, with individual waiter overrides	M
WCM-05	Optional rules by branch, order type or product category (feature-flagged)	S
WCM-06	Effective start/end dates; record of who changed a policy and when	M
WCM-07	Preview of estimated commission before activating a policy	S
WCM-08	Commission statements for daily, weekly and monthly periods	M
WCM-09	Approval and payout records, separate from customer payments; the restaurant retains control of approval and payout	M
WCM-10	The commission rate and basis are saved as a **snapshot** on each ledger entry. Changing a policy must never silently recalculate earlier earnings	M
WCM-11	Commission is always calculated on the backend; amounts supplied by the frontend are never trusted	M
WCM-12	Ledger entries are created once per eligible order and move through pending → approved → paid, or reversed	M
WCM-13	Instalment or split payments (cash + mobile money): eligibility follows policy and the amount actually paid	M
WCM-14	A refund creates a corresponding reversal or adjustment entry; the original entry is never erased	M

5.3 Waiter links and customer portal

Portal URLs (proposed patterns; adapt to existing domain and routing):

- Restaurant portal: `https://order.dukadigital.co.tz/restaurant/{slug}`

- Waiter link: `https://order.dukadigital.co.tz/restaurant/{slug}/w/{WAITERTOKEN}`

ID	Requirement	Pri
CPT-01	Each waiter has a **Share My Ordering Link** button (copy, WhatsApp, SMS, other channels) in their account	M
CPT-02	The link identifies the restaurant and the waiter; it uses a real, restaurant-scoped token, stored hashed, never a guessable code	M
CPT-03	A waiter link is **not an authentication credential**; customers need no staff password or session to order	M
CPT-04	Links can be deactivated or rotated if compromised; revoked links stop attributing orders	M
CPT-05	The portal shows that restaurant's products, prices, opening hours, available ordering methods and branding	M
CPT-06	Customer chooses products, quantities, table or delivery details, and payment method	M
CPT-07	Public endpoints expose only the minimum information needed to place and track an order	M
CPT-08	Portal works on desktop, tablet and mobile browsers, and is installable (PWA) where supported	M
CPT-09	If a customer opens the general link and later selects a waiter, the assignment is recorded explicitly. A confirmed order's waiter cannot be changed by editing the URL	M

Customer flow when using a waiter link

1. Customer opens the link; the system identifies the restaurant and waiter.
2. Customer sees the correct restaurant menu.
3. Customer places an order (products, quantities, table/delivery, payment method).

4. The system creates the order, assigns the identified waiter and marks the source (e.g. CUSTOMER_PORTAL).
5. The waiter is notified; the order appears in their order register (sound or push where supported).
6. The shared order engine sends the order to the kitchen queue with no manual re-entry.
7. Payment and commission are tracked; commission is calculated once earning conditions are met.

5.4 Order routing rules

The system must not assign every online order to a waiter arbitrarily.

Order source	Assignment rule
Waiter's unique link	Assign to the waiter identified by the valid link
QR code on a table	Assign using the table's designated waiter, if one exists
General restaurant portal	Use customer-selected waiter, table assignment or manager-configured routing
Cashier-created order	Attribute to the cashier as creator; assign a waiter separately if appropriate
Walk-in order created by waiter	Assign to that waiter
Delivery order	Assign using the configured service or delivery workflow

The original creator and subsequent waiter assignments are stored separately.

5.5 Payment flow and commission accounting

- **PAY-01 (M)** Support direct collection (waiter records cash or initiates an authorized mobile-money payment) and cashier collection (waiter requests the bill; cashier verifies and records payment).
- **PAY-02 (M)** Payment verification reconciles transaction reference, amount, payment status and outstanding balance.
- **PAY-03 (M)** Mobile money uses the provider's verified callback or transaction verification. A frontend success message alone must never mark an order paid.

- **PAY-04 (M)** Idempotency controls ensure repeated requests or payment callbacks never duplicate payments or commissions.
- **PAY-05 (M)** Order submission, payment confirmation and commission posting run in database transactions.
- **PAY-06 (M)** Recording a customer payment is distinct from paying the waiter their commission.
- **PAY-07 (M)** Cancellations, payment adjustments and commission changes keep an immutable audit history.

5.6 Entities added by this section

Entity	Important fields
waiter_profiles	id, tenant_id, branch_id, user_id, active
waiter_order_links	id, tenant_id, waiter_id, public_token_hash, active, created_at, revoked_at
orders (extended)	id, tenant_id, branch_id, order_number, source, created_by_user_id, assigned_waiter_id, served_by_user_id, table_id, status, total, created_at
order_items	order_id, product_id, quantity, unit_price, discount, tax, line_total
commission_policies	tenant_id, branch_id, waiter_id (optional), method, rate, fixed_amount, basis, effective_from, effective_until, active
commission_ledger	tenant_id, order_id, waiter_id, policy_snapshot, eligible_amount, commission_amount, status, reversal_of_id
payments	order_id, amount, method, status, collected_by_user_id, provider_reference, confirmed_at
bill_requests	order_id, requested_by_user_id, assigned_cashier_id, status, requested_at, resolved_at

Entity	Important fields
commission_payouts	waiter_id, amount, period_start, period_end, approved_by_user_id, paid_at, payment_reference
audit_logs	tenant_id, actor_user_id, action, entity_type, entity_id, reason, created_at

These are recommended entities; existing tables must not be duplicated. All queries are scoped by tenant_id and user permissions.

5.7 Manager reconciliation report

At the end of each shift or day, the manager can select a waiter and see:

- **REC-01** All orders created, assigned and served by that waiter.
- **REC-02** Orders received through the waiter's customer link.
- **REC-03** Total eligible sales and the commission rate applied to each order.
- **REC-04** Payments collected directly by the waiter versus by the cashier.
- **REC-05** Outstanding bills, partial payments, cancelled orders and refunds.
- **REC-06** Expected cash collection, actual cash handed over and any difference.
- **REC-07** Commission earned, pending approval, approved, paid and reversed.
- **REC-08** Order-level receipts, payment references and audit history.
- **REC-09** Exports for daily, weekly, monthly and custom periods.
- **REC-10** Owner can compare waiters side by side and drill down to the exact order behind every total.

5.8 Access control for this section

Manager endpoints require manager or owner permission. Waiter endpoints return only the authenticated waiter's own data. A waiter cannot see another waiter's private earnings or another tenant's orders.

6. Site Workspace Design (per outlet)

Each site gets a well-organized workspace with a left navigation, a role-aware home dashboard, a task inbox and a day timeline (Open → Service → Close).

6.1 Common layout

- **Top bar:** site name, business date, shift status, connectivity indicator, workspace switcher, notifications.
- **Home:** today's sales vs target, covers, average check, live orders, labor %, stock alerts, pending approvals.
- **Task inbox:** checklists, approvals, receiving, counts due, expiring items, maintenance tickets.
- **Quick actions:** new order, receive delivery, log waste, start count, clock in.

6.2 Role-based areas inside the Site Workspace

Area	Primary users	Contents
Front of House	Cashier, waiter, host	POS, floor plan, reservations, waitlist, online order queue, waiter order register, Share My Ordering Link, bill requests
Kitchen & Bar	Chef, cooks, bartender	KDS (including portal and waiter-link orders), prep lists, recipe cards, 86 list, waste log, temperature logs
Store & Inventory	Storekeeper	Receiving, issue, transfers, counts, expiry, requisitions
Purchasing (site)	Manager, storekeeper	Site requisitions, POs, supplier deliveries, invoice matching
Cash (site)	Manager, cashier	Shift reports, cash drops, petty cash, daily close, deposits, bill collection, commission approvals and payouts
Customers	Manager, marketing	Feedback inbox, loyalty lookups, complaints, promotions active, customer portal settings
Quality & Maintenance	Manager, chef	Checklists, audits, equipment tickets
Reports	Manager	Site dashboards, standard reports, waiter reconciliation

Area	Primary users	Contents
Settings	Manager (limited)	Printers, stations, floor plan, local price overrides, staff PINs, waiter profiles, commission policies, order-routing rules

6.3 Day lifecycle automation

1. **Opening:** checklist, float count, stock-critical alerts, roster check, prep list.
2. **Service:** live monitoring of tickets, voids, stock-outs, labor vs sales; waiter notifications for portal orders.
3. **Closing:** cash count, checklist, waste entries, end-of-day stock snapshot (key items), auto-close of tabs, waiter cash hand-over reconciliation, Z-report.
4. **Post-close:** HQ dashboard refresh, exception report to area manager.

6.4 Workspace sizing

- **Small site:** simplified single-screen workspace; modules hidden unless enabled.
- **Large site:** department sub-workspaces (Bar, Banquet, Bakery, Delivery) with separate stations and budgets.

7. Integration Requirements

- **INT-01** Public REST and GraphQL APIs, webhooks, API keys/OAuth 2.0 with rate limits.
- **INT-02** Payment gateways, card terminals, local mobile-money providers (verified callbacks, see PAY-03), QR payments.
- **INT-03** Delivery marketplaces, reservation platforms, Google Business, social ordering.
- **INT-05** Fiscal/tax authorities' e-invoicing and device APIs.
- **INT-06** Messaging: SMS gateways, WhatsApp Business API (including sharing waiter links), email.
- **INT-07** Hardware via a local agent: printers, drawers, scales, scanners, biometric devices.
- **INT-08** SSO (SAML/OIDC), directory sync for enterprises.
- **INT-09** Data export to data warehouses and BI tools.
- **INT-10** Push/web-push notification service for waiter alerts on new portal orders.

8. Non-Functional Requirements

8.1 Performance and scalability

- **NFR-01** POS action response under 300 ms locally; order send to KDS under 1 s on LAN.
- **NFR-02** Support 10,000+ sites and 50,000 concurrent users per deployment with horizontal scaling.
- **NFR-03** Reports on one site for one year in under 5 s; group dashboards under 10 s using pre-aggregated data.

8.2 Availability and reliability

- **NFR-04** 99.9% cloud uptime (99.95% for Enterprise); RPO ⩽ 5 min, RTO ⩽ 1 h.
- **NFR-05** Offline-first local store at each site with automatic, idempotent sync and conflict resolution rules.
- **NFR-06** Optional on-site server node (site hub) so devices keep working together when internet is down.

8.3 Security and privacy

- **NFR-07** TLS 1.2+ in transit; AES-256 at rest; secrets in managed vault.
- **NFR-08** MFA for managers and above; PIN/biometric for floor staff; session timeouts.
- **NFR-09** PCI DSS scope reduction via tokenized card handling; never store card numbers.
- **NFR-10** GDPR/local data-protection compliance: consent, export, deletion, data residency options.
- **NFR-11** Tenant isolation at data and encryption-key level; periodic penetration tests.
- **NFR-12** Immutable audit trail and fraud controls (void/discount anomaly alerts).
- **NFR-21** Waiter link tokens are stored hashed, can be rotated or revoked, and public endpoints are rate-limited against enumeration and abuse.

8.4 Usability

- **NFR-13** New cashier trained in under 2 hours; touch-first UI, large targets, high contrast for bright kitchens.
- **NFR-14** Multi-language UI, RTL support, accessibility (WCAG 2.1 AA for web/customer apps).
- **NFR-15** Guided onboarding wizards, in-app help, and demo data mode.

8.5 Maintainability and extensibility

- **NFR-16** Modular services, feature flags, zero-downtime releases, backward-compatible APIs (versioned).
- **NFR-17** Configurable fields, custom reports, and workflow rules without code changes.
- **NFR-18** Automated test coverage ⩾ 80% on core financial and inventory logic, and on commission, payment idempotency and order-attribution logic.

8.6 Data and retention

- **NFR-19** Configurable retention per country law (typically 5–10 years for financial data); archival of closed sites.
- **NFR-20** Daily encrypted backups with tested restores.

9. System Architecture

9.1 Logical architecture

- **Client layer:** PWA/web, native tablet POS, handheld app, KDS, customer app and portal, supplier portal.
- **Edge layer (per site):** optional Site Hub – local database, print server, device agent, sync engine.
- **API gateway:** authentication, tenancy resolution, rate limiting, routing.
- **Domain services:** Sales, Menu, Inventory, Procurement, Production, CRM, Reporting, Workflow, Notification, Integration, Commission.
- **Data layer:** transactional database (PostgreSQL), cache (Redis), event bus (Kafka or equivalent), object storage for files, analytics warehouse (columnar) for BI.
- **Platform services:** identity and RBAC, audit, configuration, localization, job scheduler, search.

9.2 Key architectural decisions

1. Modular monolith first, services where needed for small deployments; extract high-load services (Sales, Reporting) as scale grows.
2. Event-driven: every sale, receipt, waste or adjustment emits an event; inventory, finance, loyalty, commission and analytics subscribe.

3. Multi-tenancy: shared infrastructure with tenant ID row-level security; dedicated database option for large enterprises.
4. Offline-first sync: local event log with unique IDs, replay to cloud, deterministic conflict rules (sales never overwritten; master data last-approved-wins).
5. CQRS for reporting: read models and aggregates per site/day so HQ dashboards never slow down POS.
6. Configuration over customization: site templates, rules engine and workflow designer.
7. One shared order engine serves POS, waiter handheld, QR, customer portal and waiter links; no separate ordering stacks per interface.

9.3 Deployment options

Multi-tenant cloud (default), single-tenant cloud, hybrid (cloud plus site hub), on-premise (rare, enterprise). Containerized (Kubernetes), infrastructure as code, multi-region with regional data residency.

10. Core Data Model (summary)

- **Organization:** Tenant, LegalEntity, Brand, Region, Site, Station, User, Role, Permission, WorkspaceMembership.
- **Catalog:** Item, ItemVariant, Modifier, Combo, Recipe, RecipeLine, PriceList, PriceRule, TaxRule, Allergen.
- **Sales:** Order, OrderLine, Check, Payment, Refund, Shift, CashMovement, Reservation, Table, Customer, LoyaltyAccount, BillRequest.
- **Waiter and commission:** WaiterProfile, WaiterOrderLink, CommissionPolicy, CommissionLedger, CommissionPayout.
- **Inventory:** StockItem, Location, StockLedger (immutable movements), Batch, Count, CountLine, Transfer, WasteEntry.
- **Procurement and Production:** Supplier, Requisition, PurchaseOrder, GoodsReceipt, SupplierInvoice, ProductionOrder, DispatchNote.
- **System:** AuditLog, WorkflowDefinition, ApprovalRequest, Notification, IntegrationEndpoint, SyncQueue.

Principle: inventory and commission use append-only ledgers; corrections are new reversing entries, never edits.

11. Key Business Rules

1. A sale posts revenue, tax, payment, theoretical stock depletion and COGS in one atomic event chain.
2. Variance = actual stock minus theoretical stock; variances above threshold require manager reason and approval.
3. Discounts, voids and refunds above role limits require supervisor authorization and are always logged with reason.
4. A closed business day cannot be edited; adjustments are posted in the current open period with reference.
5. Purchase orders above approval limits cannot be sent to suppliers until approved.
6. Items expiring within configured days are blocked from sale/use unless overridden with a reason.
7. Price and menu changes at HQ propagate to sites on a scheduled effective date unless a site override exists.
8. Franchise royalties are computed on net sales per contract and invoiced automatically.
9. Commission is posted once per eligible order, calculated on the backend from the policy snapshot; policy changes never recalculate posted entries.
10. A refund reverses commission through a new reversing entry; the original entry is retained.
11. A waiter link attributes an order but never authenticates anyone; a confirmed order's waiter cannot be changed by editing a URL.
12. A frontend payment success message never marks an order paid; only a verified provider callback or an authorized staff record does.

12. Reports and KPIs

Area	KPIs
Sales	Revenue, covers, average check, channel mix, table turn, sales per waiter, share of orders via waiter links
Cost	Food cost %, beverage cost %, waste %, variance %, supplier price index
Customer	Repeat rate, NPS/rating, loyalty share, delivery time

Area	KPIs
Control	Voids %, discounts %, refunds, cash over/short, audit scores, waiter cash hand-over differences
Waiter commission	Commission earned per waiter, commission cost % of sales, commission liabilities (pending, approved, paid)

13. Implementation and Rollout Plan

13.1 Phases

Phase	Duration	Deliverables
1. Foundation	3–4 months	Tenancy, org hierarchy, RBAC, workspace engine, menu, POS, KDS, payments, basic reports, offline sync, shared order engine
2. Operations	3 months	Inventory, recipes and costing, procurement, waste, counts, shift/cash management, day-close, commission policies and ledger, unique waiter links and automatic order attribution
3. Business	3 months	tax and fiscal, CRM/loyalty, reservations, online ordering and customer portal, payment verification, manager reconciliation
4. Scale	3–4 months	Multi-site HQ, central kitchen, franchise, benchmarking, approval designer, integrations marketplace
5. Intelligence	ongoing	Forecasting, AI insights, anomaly detection, IoT, advanced BI

Recommended order for the waiter/portal capability: (1) commission policies and ledger; (2) unique waiter links and automatic attribution; (3) connect existing kitchen and payment flows; (4) manager reconciliation and automated tests.

13.2 Customer onboarding by size

- **Small:** self-service setup wizard, template menu import, go live in 1 day.
- **Medium:** guided implementation (2–6 weeks): data migration, hardware setup, training per workspace role, pilot site, then roll-out.
- **Large:** program approach (3–9 months): discovery, fit-gap, integrations, pilot cluster, wave roll-out, hypercare and center of excellence.

13.3 Training and support

Role-based training paths, in-app tutorials, sandbox environment, 24/7 support for Enterprise, status page and SLA reporting.

14. Acceptance Criteria

14.1 Platform

1. A new site can be created from a template and take its first order within 1 business day (Starter) / 1 hour after hardware arrives (Growth+ clone).
2. POS continues all core functions with the internet disconnected for 72 hours and syncs without duplicate or lost transactions.
3. A sale correctly updates stock by recipe, appears in HQ dashboards within 60 seconds of connectivity.
4. Users can see only the workspaces and data granted to them; cross-tenant access tests all fail.
5. Day-end close produces Z-report, cash reconciliation with zero manual re-keying.
6. Pilot customers achieve measurable improvement: food-cost variance reduced and reporting time cut within 90 days.

14.2 Waiter, commission and customer portal

7. A manager can configure and activate commission rules, including a waiter-specific override.
8. Changing a policy does not silently alter previously posted commission entries.
9. Each waiter can copy and share their own customer portal link.
10. An order placed through that link is attributed to the correct waiter and restaurant.
11. A general portal order follows the restaurant's configured assignment rules.

12. An order appears in the waiter register and kitchen queue without duplicate creation.
13. The waiter can collect an authorized payment or request a cashier bill.
14. Repeated payment callbacks cannot duplicate payments or commissions.
15. A waiter cannot access another waiter's private earnings or another tenant's orders.
16. A manager can reconcile orders, payment collections, outstanding bills and commissions by waiter, branch and period.
17. Cancellations, refunds and commission payouts preserve an auditable record.
18. The system works on desktop, tablet and mobile browsers, including the installable customer portal where supported.

15. Risks and Mitigations

Risk	Mitigation
Poor connectivity at sites	Offline-first design, site hub, lightweight clients
Complexity overwhelming small restaurants	Tiering, feature flags, simplified starter workspace, wizards
Data quality (recipes, units, opening stock)	Import validation, guided counts, templates
Hardware diversity	Certified hardware list, universal device agent
Regulatory differences	Localization packs, configurable tax/fiscal engine, local partners
Staff resistance	Role-based training, simple UI, quick wins (faster service, less paperwork)
Fraud and shrinkage	Audit trail, approval limits, anomaly alerts, blind counts
Scope creep	Modular roadmap, prioritization by tier and customer value
Waiter link abuse or leakage	Hashed tokens, rotation/revocation, rate limits, link never authenticates
Disputes over commission	Policy snapshots, append-only ledger, audit history, preview before activation

Risk	Mitigation
Duplicate payments or commissions	Idempotency keys, transactions, verified provider callbacks

16. Traceability and Next Steps

Each requirement ID (POS-xx, INV-xx, WCM-xx, CPT-xx, PAY-xx, REC-xx, etc.) maps to user stories, test cases and release milestones.

Next steps: (1) validate priorities with 3–5 pilot restaurants (one small, one medium, one chain); (2) produce UX prototypes of the Site Workspace, HQ Workspace and Customer Portal; (3) finalize technical architecture and sync design; (4) build the Phase 1 backlog and estimate; (5) select localization targets (tax, fiscal, payments) for the first launch markets.

Appendix A – Example Permission Matrix (excerpt)

Action	Cashier	Waiter	Chef	Storekeeper	Site Mgr	Area Mgr	Owner
Take order	✔	✔	–	–	✔	–	–
Void item	Limit	Limit	–	–	✔	✔	✔
Apply discount	Limit	–	–	–	✔	✔	✔
Receive goods	–	–	–	✔	✔	–	–
Approve PO	–	–	–	–	Limit	✔	✔
Edit recipe	–	–	Propose	–	Propose	✔	✔
Close day	–	–	–	–	✔	–	–
View all sites	–	–	–	–	–	Region	✔
Share own ordering link / view own orders and earnings	–	✔	–	–	–	–	–
Record payment / handle bill request	✔	Limit	–	–	✔	–	✔
Configure commission policy	–	–	–	–	✔	Region	✔
Approve commission payout	–	–	–	–	Limit	✔	✔
Waiter reconciliation report	–	–	–	–	✔	✔	✔

Appendix B – Typical Site Day Flow

Open → Checklist → Cash float → Prep list → Service (orders incl. portal and waiter-link orders, KDS, deliveries, bill requests) → Receiving → Waste logging → Close → Cash count and waiter hand-over reconciliation → Z-report → Auto-posting to finance → HQ summary.

Appendix C – Suggested API Endpoints (waiter, portal and commission)

Adapt to existing FastAPI router conventions and API version; do not duplicate existing functionality.

Endpoint	Purpose
GET /api/v1/waiter/me/orders	Waiter's own order register

Endpoint	Purpose
GET /api/v1/waiter/me/earnings	Waiter's commission ledger and payout history
GET /api/v1/waiter/me/order-link	Retrieve the waiter's active sharing link
POST /api/v1/waiter/me/order-link/rotate	Rotate or replace the sharing token
GET /api/v1/restaurant/{slug}/portal	Public restaurant portal configuration
GET /api/v1/restaurant/{slug}/menu	Public menu for that restaurant
POST /api/v1/restaurant/{slug}/orders	Create customer portal order
POST /api/v1/orders/{order_id}/request-bill	Submit a bill request
POST /api/v1/orders/{order_id}/payments	Record or initiate a permitted payment
GET /api/v1/manager/waiters/reconciliation	Cross-check all waiters and orders
GET /api/v1/manager/commissions/policies	List commission policies
POST /api/v1/manager/commissions/policies	Create or configure a commission policy
POST /api/v1/manager/commissions/payouts	Record an authorized commission payout

Manager endpoints enforce manager or owner permissions; waiter endpoints enforce access to the authenticated waiter's own data; public endpoints expose only the minimum needed to place and track orders.